

Onotsavam Quiz Buzzer

Setup guide for the host

Teams buzz in from their own phones, and you get the full order they buzzed in.

Everyone playing just opens a web page on their phone — nothing to download, no app, no accounts. One device has to run the quiz itself, and that is all this guide sets up. Do it once in advance, not on the day.

Start here: which device runs the quiz?

SETUP A

An Android phone

Use this if the laptop is a work machine that blocks installing software, or will not let other phones connect to it.

The phone shares the hotspot and runs the quiz. The laptop only opens a web page, which no work policy blocks.

→ **Go to Setup A, then Running the quiz**

SETUP B

A laptop

Use this if you can install software on the laptop and click Allow when Windows asks about the firewall.

Simplest option on a personal laptop. On a locked-down work laptop it usually fails, and you cannot fix it without an administrator.

→ **Skip to Setup B, then Running the quiz**

Not sure? If it is your own laptop, use Setup B. If it belongs to your employer, use Setup A — a work laptop normally refuses both the install and the incoming connections from phones, and neither can be worked around without admin rights.

Running it from an Android phone

One Android phone does everything: it shares the hotspot, runs the quiz, and shows your quizmaster controls. Here is who does what:

Android phone	Hotspot + the quiz itself + your private controls. Sits on a charger.
Laptop	Joins the hotspot and opens the big screen page. Nothing installed.
Team phones	Join the hotspot and open the buzzer page.

Do this the day before. Downloading Python onto the phone takes a few minutes and needs internet. Plug the phone into a charger — a hotspot and a server together drain a battery fast.

1 Install Termux on the phone

Termux is a free terminal app. Open this page in the phone's browser:

github.com/termux/termux-app/releases

Open **Assets** under the newest release and download the file ending **universal.apk**. Android will warn about installing from an unknown source — allow it for your browser. That is a normal phone setting and needs no special permission.

Do not install Termux from the Play Store. That version is out of date and step 3 will fail on it, with an error that does not explain why.

2 Open Termux and update it

You get a black screen with a prompt. Type this and press Enter:

```
pkg update -y && pkg upgrade -y
```

If it asks anything, press Enter to accept. Takes a couple of minutes.

3 Install Python

```
pkg install -y python git
```

Check it worked with `python --version` — it should print a version number.

4 Download the quiz

```
git clone https://github.com/events26/Fest_26_Onam.git
```

```
cd Fest_26_Onam
```

That is everything downloaded. You will not need internet again except to sign in as quizmaster.

5 Turn the hotspot on — before starting the quiz

Leave Termux open, go to Settings, turn on the phone's hotspot, then switch back to Termux.

Order matters here. The quiz works out its address the moment it starts, so if the hotspot is off it will print an address nobody can reach.

6 Stop Android from shutting it down

```
termux-wake-lock
```

Do not skip this. Without it Android suspends Termux when the screen sleeps, and every phone in the room simply stops responding mid-question with nothing on screen to explain it. Also worth setting Termux to **Unrestricted** battery usage in Android Settings.

7 Start the quiz

```
python quiz-server.py
```

It prints the addresses. On an Android hotspot the join address is almost always:

```
http://192.168.43.1:8000/
```

If it printed something different, use what it printed. If it listed more than one, try them in order.

Leave Termux open and the screen on. You can dim it, but do not close the app. Now skip to **Running the quiz**.

1 Download the code

https://github.com/events26/Fest_26_Onam

Click the green **Code** button, then **Download ZIP**. You do not need a GitHub account or any other program.

2 Unzip it

Right-click the downloaded file and choose **Extract All...**, then **Extract**. You get a folder called `Fest_26_Onam-main`. Keep it somewhere easy, like the Desktop.

On a Mac — just double-click the ZIP, it unpacks itself.

3 Install Python

Go to python.org/downloads and click the big yellow **Download Python** button, then run the file.

This is the one step people get wrong. On the installer's first screen, tick **"Add python.exe to PATH"** at the bottom *before* clicking Install Now. Miss it and nothing later works, and the error will not tell you why.

Cannot install anything on this laptop? Open **NO-ADMIN-SETUP.txt** in the quiz folder — there is a portable Python that is just a ZIP, no installer and no admin needed. If phones then still cannot reach the laptop, use **Setup A** instead; that is a firewall restriction you cannot remove without an administrator.

On a Mac — no PATH box to worry about. Open Terminal ( **Space**, type Terminal) and run `python3 --version`. If it offers to install developer tools, click Install.

4 Test it

Open the folder and double-click `start-quiz.bat`. A black window should open showing three web addresses. Leave it open — closing it stops the quiz.

On a Mac — use `start-quiz.command`, and the **first** time right-click it and choose **Open** rather than double-clicking. macOS blocks downloaded scripts, and a double-click just refuses with no way forward. After that once, double-clicking works.

5 Allow it through the firewall

Windows asks whether to let Python communicate on the network. Click **Allow access** with **Private networks** ticked.

Click Cancel and phones will never connect, with no other clue what went wrong. If the box never appeared, or clicking it asked for an administrator password, this laptop cannot host the quiz — use **Setup A** instead.

6 Check it from a phone

In the black window find the line starting **Teams join**. It shows an address like `http://192.168.1.42:8000/` — the numbers differ on every laptop, so use whatever *your* window shows.

Put one phone on the **same Wi-Fi as the laptop** and open that address. If the team page loads, you are ready.

Running the quiz

1 Get everyone on one network

Every phone and the laptop must be on the **same hotspot** — the phone's own hotspot in Setup A, or whatever the laptop is on in Setup B.

Avoid office, hotel and guest Wi-Fi. These stop devices talking to each other, so phones will never reach the quiz no matter what else you do. A personal hotspot avoids it completely.

2 Put the big screen up

On the laptop, join the hotspot and open the join address with `/display` on the end — for example `http://192.168.43.1:8000/display` — then connect the projector and:

- Click **Full screen**
- Click **Turn on sound** — browsers block sound until you click once, so this has to happen before the first question

3 Open your own controls

Open the join address with `/admin` on the end, then sign in.

Keep this on a phone, not the projected screen. It shows the answers. In Setup A you can use the server phone itself at `http://127.0.0.1:8000/admin`.

Ask Sanjay for the username and password before the day. There is no separate quiz password — it is the same login as the main Onam admin page, so you cannot get in without it.

Signing in needs internet, so do it early. Once you are in you stay in, even if the connection drops or the quiz restarts.

4 Set up the teams

Open **Teams & codes**. Change the team names if you need to, then tap **Generate codes** and **Save teams & codes**.

Read each code out to **one person per team**. That phone holds the buzzer for the whole quiz.

5 Let the teams in

Read out the join address. Each team opens it, picks their team name and types their code.

Watch the projector: each team's **NOT JOINED** label disappears as they sign in, so you can see who is still missing without asking.

6 Run each question

Read the question aloud from your phone, then tap **Arm buzzers**. Every phone lights up and starts pulsing. Teams tap to buzz. The first one takes the floor and the big screen fills with their colour — and **everyone who buzzes after them lines up behind them**, in the order they tapped. Your phone shows the whole running order:

1	Maveli Squad	0.41s	ANSWERING
2	Gajaveerans	0.63s	
3	Vallam Vikings	0.88s	
4	Onam Kombans	1.12s	

Then:

- **Correct** gives them 10 points and closes the question
- **Wrong — next team** crosses them off and passes the floor to number 2, then 3, and so on down the list
- Or **tap any team in the list** to jump straight to them
- **Show the answer on screen** highlights the right option on the projector
- **Next question**

The buzzers stay open all the way through. A team that hesitated can still tap and join the end of the queue. If everyone who buzzed gets it wrong, the big screen says so and keeps waiting.

Reset this question starts one over completely, for when a question has to be thrown out.

If something goes wrong

WHAT YOU SEE	WHAT TO DO
Setup A: “pkg: command not found”	Termux came from the Play Store. Uninstall it and install the APK from the releases page in step 1.
Setup A: everything freezes when the phone screen goes off	<code>termux-wake-lock</code> was not run, or Android is still restricting the app. Run it, and set Termux to Unrestricted battery usage in Android Settings.
Setup A: Termux closed by accident	Reopen it and run <code>cd Fest_26_Onam</code> , then <code>termux-wake-lock</code> , then <code>python quiz-server.py</code> . Team names, codes and scores are all saved — teams just open the join page again.
Setup A: the address does not load on other phones	Check the hotspot is on and everyone is connected to it, not the office wifi. Then press Ctrl-C, check the hotspot again, and start the quiz once more so it re-reads its address.
Setup B: the black window says Python was not found	If you installed Python, the Add python.exe to PATH box was missed — reinstall and tick it. If you cannot install anything, see NO-ADMIN-SETUP.txt .
Setup B: phones cannot open the page, but the laptop can	The firewall is blocking it. Right-click Start, choose Terminal (Admin) and paste this in: <code>New-NetFirewallRule -DisplayName "Onam Quiz" -LocalPort 8000 -Protocol TCP -Action Allow</code> If that needs a password you do not have, switch to Setup A .
Setup B: “start-quiz.command cannot be opened”	On a Mac, right-click the file and choose Open , then Open again on the warning. Only needed once.
Cannot sign in to the admin page	Signing in needs internet. Check that phone can load any website, then try again.
A team's phone died or was locked out	They sign in again with the same code — the newest phone takes over the buzzer. You can also tap Sign out next to their name on the admin page.
Everyone buzzes instantly, just to grab a place in the order	Make a wrong buzz cost something. Open <code>quiz-server.py</code> , find <code>POINTS_WRONG = 0</code> near the top, change it to <code>POINTS_WRONG = -5</code> and restart. Guessing usually stops within a round.
No sound from the projector	Turn on sound was not clicked on the display page. Move the mouse to make the buttons appear, then click it.

Worth a five-minute rehearsal. Run through the last section with two phones the day before. Everything that goes wrong on the day goes wrong in the first two minutes, and it is far easier to fix when there is not a room full of people waiting.

